eego™ Software Release Notes

Version number	***1.8.2***

The software is pre-installed on your system. Before you can start working you must agree to our terms and conditions as presented on the first screen in the **eego**™ software.

Changes and History

Version 1.8.2	Internal issue ID
New Features since version 1.8.0	
Allow the user to define a stream-specific network event offset for LSL event streams	4533
Bug fixes since version 1.8.0	
Impedance values slightly unstable for caps with Sn electrodes.	2787
Sensor data of upgraded database cannot be reviewed	3853
Wrong DisplayUnit in montage after DB update: no recording possible.	3921
Setting ‘EventsForAllAmplifiers’ was inverted, leading to triggers from all amps in cascaded set-up when setting was ‘false’ and vice versa.	4009
GSR sensor value export faulty if export was executed several times.	4182
Improved stablity regarding sample loss during recording. Note: on some PC hardware, there is still the chance of loosing some USB data packages after updating to Win10 1710 or above. The software will mark such events as "Amplifier disconnect / Amplifier reconnect" and start a new recording segment automatically.	4683
Corrected timestamps of new segment after sample loss	4684
Update of database with incomplete video recording failed	4696
EEG display delayed up to a second	
Limitations	
Bipolar channels only support sampling rates up to 8192Hz, so to record EEG with additional bipolar channel or sensor information, choose a sampling rate up to 8192Hz. If a higher sampling rate is chosen, bipolar and sensor channels will deliver flat lines (0V).	2100
Video recording properties like format, file size, resolution cannot yet be set.	2284
Playback speed of video cannot be changed while the video is playing. Press pause, change speed, press start again.	3124
Channel-specific events are exported to file without channel information, i.e. they appear as global events in 3rd party software.	3567
Export or review data on a 32bit operating system sometimes fails due to memory limitations for 32 bit applications. Workaround: Try again. Internally, eego will then try to reduce memory footprint, but might need longer for the operations to complete. Note that we strongly encurage you to use the 64 bit version of eego.	3816
The artefact reduction module for recordings with dry electrode caps can only be used for online data correction for acquisition with up to 4096Hz. Note that this restriction does not affect offline correction: In review, the correction is supported for of all sampling frequencies.	3823
Export to Neuroscan format: triggers only exported if export contains only one segment. Workaround: Export to EEProbe cnt format or EDF+ format.	3825
Known problems	

After a restart of the computer, very rarely no data are delivered from the amplifer because of a USB error. In such a case, disconnect and re-connect the USB cable.	2612
After an error, sometimes the amplifer state or battery state are displayed incorrectly in the status bar. Restarting eego and the amplifier helps.	2624, 2625, 3786
Very rarely, the amplifier cannot switch into impedance mode. In such a case, restart the amplifier and the recording.	2702
Archiving of recordings with averages not yet implemented.	2705
BrainVision export does not set reference channel label	3099
Exporter: eego closes when storage device is disconnected	3255
Export: eego stops working when there is not enough space on storage device	3256
BrainVision export results in unreadable file for file size bigger than 2GB	3279
Sometimes, some flashes are delayed from the Photic device	3282
Limited anti-aliasing functionality in EEG review.	3489
Sometimes, video camera cannot be started. Workaround: Try starting again or disconnect & reconnect device. Try closing applications that also use the camera, e.g. Skype.	3577
Multisampling for EEG display not working.	3695
Sometimes, archiving or deleting a recording directly after the recording or review does not work. Workaround: Try to restart eego and execute delete or archive without starting another workflow first.	3721
Events loose their "expanded" state in review if user starts scrolling.	3781
Event ticks in the scrollbar can be displaced or missing in online review.	3810
Deleted events still appear in the scrollbar.	3813
If recording is continued after an unavailable amplifier becomes available again, an additional segment of a few samples in length is inserted in the recording.	3815
In the Edit amplifier workflow, it is not possible to change the channel type of a newly created channel if the last channel in the list is not a bipolar channel. Workaround: Use move down a bipoar channel until it is the last. Create a new channel of desired type. Move the bipolar channel up again.	3819
If the name of a sensor is changed in the hardware setup after creating a related montage, the original montage cannot be used any more for recordings with this hardware setup or reviewing recordings with the changed hardware setup. Workaround: Delete the montage, create a new one.	4235
Sometimes, video cannot be started. Workaround: restart eego and try again.	4282
EEG shown is to 'short' (not filling the whole screen) if display time spans below 10ms are selected	4584

Version 1.8.0	Internal issue ID
New Features since version 1.7.2	
Support for eego amplifier series EE-4xx.	
Artefact reduction module for recordings with dry electrode caps: method for reducing the artefacts, resulting in 64 cleaned channels, and interpolation method that projects the 64 channel data to a standard 21 channel clinical layout. Feature is for research-only use and requires special license!	
New montage concept: Montages are not bound to amplifier set-ups any more. Any montage can be selected to inspect EEG data. If montage channels are defined that are unavailable from the amplifier set-up, those channels will automatically be hidden from view.	
Define manual annotations: Users can define up to 1000 different annotation types. Annotations can have fixed or variable length, user-defined colors, user-defined text and user-defined code in the range from 1000 to 1999. Manual annotations can be used during EEG acquisition and EEG review. Annotations can be channel-specific or global.	
Edit annotations in EEG review. Manual annotations can be edited when reviewing EEG recordings: Users can change start, duration, text and code. Users can delete manual annotations. Editing annotations in render window and events grid are supported.	
Edit amplifier set-up workflow: Divided workflow in two steps: select amplifier set-up and edit amplifier set-up. New options to "save", "save as" or "Cancel" when editing an amplifier set-up.	2403

Option to hide an amplifier set-up from list of available amplifier set-ups in acquisition workflow.	
Edit montage workflow: Divided workflow in two steps: select montage and edit montage. New options to "save", "save as" or "Cancel" when editing a montage. Option to hide a montage from list of available montages in acquisition workflow or review workflow.	2403
Support for external triggers, streamed via the LSL interface. Feature is for research-only use and requires special license!	
Support EDF+ data format for export, replaces EDF format.	
Better performance for recordings with many triggers.	
Dynamic adaptation of render framerate to support PCs with lower performance.	

Bug fixes since version 1.7.2

On rare occurances, rendering stops when starting a remote desktop connection. Recording will still continue in this case.	2104
It is possible to save a Hardware Setup with all channels disabled	2805
256 channel layout for impedance slightly distorted	2815
Some sensor units exported to cnt (eeprobe format) are not available in asa 4.9.3. Solution: Update to asa 4.10.1 or above.	3027
Sometimes, video is not centered in the window or the video window is too small. Workaround: reconnect camera and / or restart eego acquisition workflow.	3044, 3155
Sometimes, video cannot be started. Workaround: reconnect camera and / or restart eego acquisition workflow.	3066, 3090
Sometimes, video shows unstable behaviour.	3131, 3264
No warning that a exported file is overwritten when exporting more than one recording or exporting to EDF	3149, 3260
Sometimes, a recording cannot be deleted because of a timeout of the database. Workaround: Start eego again.	3159
Clicking the "Subject" button in the Edit Subject workflow makes a newly created subject disappear.	3271
Duke 128 channel impedance layout wrong	3300
Exception when attempting to review recordings containing segments with zero length	3330
Unified time representation to 24 hours time format	3353
Start/ stop video switches the focus always to online review instead of staying in the current view	3654
If user settings config file is corrupt, eego does not start	3750
Channel selection in impedance mode not correctly synchronized for electrodes arranged in a grid pattern.	3761

Changes since version 1.7.2

Updated database format to 7.1.6. Old database versions will be updated automatically when connecting eego 1.8.0 to them for the first time.

List of pre-installed default montages now layout-specific ("10-5 layout", "equidistant layout") and not cap specific ("CA-xxx") any more. Users should select the 10-5 layout montage for CA-xxx caps and the equidistant layout montage for CY-xxx caps.

Changed representation of events in rendering. They are now drawn only at the bottom of the page. Click or tab on them to extend them vertically over the whole screen.

To support remote access to eego, enable the "Remote Desktop Protocol compatible display" setting in the Application Options -> General section. This switches the EEG display to software rendering and allows the EEG traces to be transmitted via the remote desktop protocol.

New installer technology

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Known problems

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Version 1.7.2	Internal issue ID

New Features since version 1.7.1

Databases can be created in user-defined folders. This allows to have databases on additional drives of the computer, e.g. an extra hard drive in a desktop PC or an SD card in a tablet PC.

Added support for EE-430 amplifier

Support for Dell tablet Venue Pro 5855

Bug fixes since version 1.7.1

Channel selections from EEG preview mode are not synchronized with the channel selections in Impedance mode.	2756
Occasionally empty EEG display after rapid montage changes	3190
Default montage has a star that marks it as edit, when it is initialized the first time	3195
Wrong focus behavior of ribbon bar when switching between EEG acquisition and EEG online review tabs	3202
In review: Filter transient response visible on each page when having a higher sampling rate and a low highpass filter value	3216
EEG export to 3rd party file formats might fail on PCs with low performance	3232
Impedance values are not exported correctly to cnt	3236
Exception while adding channels to the montage with copy / paste	3267
In tablet mode, buttons in Delete Amplifier and Delete Montage dialogue partially occluded	3269
In acquire: video windows is white after shrink and increase	3270
Recording cannot be deleted sometimes due to a "reference constraint in the 'FK_Videos_PathLocator_VideoStore_path_locator'". We now prevent that such recordings enter the database. If you have such a recording comming from an older eego version, please contact support if you want to have it deleted.	3272
Prevent empty annotation strings in the options	3281
Manual events displayed incorrectly in Events Grid	3282
Under rare and special circumstances, the amplifier does not deliver data with the commanded sampling rate, but with 500Hz after switching back from impedance mode to EEG mode.	3287
In review, an event on the last sample of segment is not displayed if the segment is not the last segment	

Changes since version 1.7.1

Highpass filter value of 0.01Hz added to available filter values.	3234
Now showing "unlimited" instead of 9999-12-31 for an unlimited license	3252
EEG acquisition can be continued after the USB connection to the amplifier has been lost or the amplifier has been switched off as soon as the amplifier is connected and running again. Recording will continue with a new EEG segment.	

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Video recording properties like format, file size, resolution cannot yet be set.	2284
Playback speed of video cannot be changed while the video is playing. Press pause, change speed, press start again.	3124

Known problems

On rare occurances, rendering stops when starting a remote desktop connection. Recording will still continue in this case.	2104
After a restart of the computer, very rarely no data are delivered from the amplifer because of a USB error. In such a case, disconnect and re-connect the USB cable.	2612

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It is possible to save a Hardware Setup with all channels disabled	2805
256 channel layout for impedance slightly distorted	2815
Some sensor units exported to cnt (eeprobe format) are not available in asa 4.9.3.	3027
Sometimes, video is not centered in the window or the video window is too small. Workaround: reconnect camera and / or restart eego acquisition workflow.	3044, 3155
Sometimes, video cannot be started. Workaround: reconnect camera and / or restart eego acquisition workflow.	3066, 3090
BrainVision export does not set reference channel label	3099
Sometimes, video shows unstable behaviour.	3131, 3264
Sometimes, a recording cannot be deleted because of a timeout of the database. Workaround: Start eego again.	3159
Exporter: eego closes when storage device is disconnected	3255
Export: eego stops working when there is not enough space on storage device	3256
No warning that a exported file is overwritten when exporting more than one recording or exporting to EDF	3149, 3260
Clicking the "Subject" button in the Edit Subject workflow makes a newly created subject disappear.	3271
BrainVision export results in unreadable file for file size bigger than 2GB	3279
Sometimes, some flashes are missing from the Photic device	3282

Version 1.7.1	Internal issue ID
New Features since version 1.6.2	
Review of the recorded EEG already during recording ("Online review")	
True 64bit application, able to handle more than 2GB of data (depending on available RAM on the PC)	
Possibility to insert annotations in review (online and offline).	2963
Annotations can have a duration now.	2828, 2967
Free-text annotations: at the click of a button, a dialog appears and any text entered there will be placed in the data as an annotation at the time when the button was pressed	2962, 3045
Digital zoom and panning for video	
Option to save a montage during Acquisition or Review	
Improved Review performance by storing and loading a downsampled version of the EEG for sampling rates above 4kHz.	
Improved GUI	
Possibility to recover the full EEG after an unexpected termination of the recording, e.g. due to a power shut-down of the PC	
Improved EEG rendering	
Improved annotation display: annotations are now displayed on the bottom of the EEG view, so they do not block the entire EEG view in case there are a high number of events.	
Touch support and arrow key support for scrolling through the EEG in review	2885
Extended default amplifier set-ups: 1) with or without bipolar channels; 2) named according to matching waveguard cap or generic; 3) headbox or tyco adapters.	

Several channels with different sensitivity can now be scaled up or down at the same time by selecting them and pressing the "up" / "down" sensitivity controls in the ribbon.	2811, 2865
Montage changes in acquisition and review are now marked with a * next to the montage name in the combobox	2958
Extended status bar in review: display patient, recording and segment information as well as archive information	3091
Extended tooltip for amplifier information	2952
Floating windows (e.g. video) are now indicated and selectable via the windows task bar	2964
Improved installer, uninstalls previous eego versions automatically and sets up the SQL database	2910, 2994, 3021

Bug fixes since version 1.6.2

Visibility in the subject editor does not work as expected	2216
Eego freezes if connection with database is interrupted while recording.	2271
Data is lost if connection with database is interrupted while recording. There are only extreme cases that lead to a loss of database connection, e.g. forcefully kill the eego application. For extended safety, users might enable the cnt export to file during acquisition.	2272
Duration of video recording not yet visualized in review.	2311
In a video session with 1 monitor after a video session with two monitors, the undocked video window can be out of the bounds of the monitor.	2396
Montage editor: move ref channel past bipolar channel duplicates bipolar channel in view	2499
Channels are not centered after montage change	2534
Sometimes, amplifier state is displayed wrongly (power on / off), sometimes, the battery state is not presented althouh the amplifier is on.	2624, 2634
Negative Kelvin do not exist by definition	2712
On rare occurances, eego might fail with an OutOfMemory exception. The software tries to change the configuration in that case to minimize memory consumption in trade off for computation speed. Fixed for 64bit version of eego.	2774, 2775
Sometimes, USB detach of a video camera during video recording causes eego to shut down.	2776, 2926
Scrolling in Impedance Mode just partly possible	2800
Amplifier battery level status bar indication not matching real amplifier battery status indication.	2827
Sometimes, a trigger is not rendered during acquisition	2839, 3009
Wrong behaviour of Montage and Amplifier editor grids	2841
Starting video stopped eego	2975
Sometimes, video recording could not be started	2976
Export of sensor data of units kiloohm or permille to cnt gives only flat lines	2979
Bipolar channels with inverted amplitudes	2991
Text truncated in tablet GUI	2998
Color of uninitialized Impedance for GND and Ref is different in comparison to the other electrodes	2999
Combobox "channels on screen" should have values adjusted for selected montage	3049
In montage setup: it is possible to set more then one montage as default	3058
Sometimes, streaming doesn't start for cascaded amplifiers	3059
Reference same as channel should give flat line	3075
Exceptions in filtering	3103, 3151
Closing the application during a running recording could lead to segments of lengths zero in the database, making the data unaccessible.	3132
No warning if an edf file gets overwritten in export.	3149

Changes since version 1.6.2

eego amplifiers EE-201, EE-202, EE-203, EE-206, EE-207 not supported any more.	
Channels that are set to invisible in the montage are not exported to file any more.	
Downsampling for render purposes now uses averaging instead of subsampling	
Video window always starts undocked	2956
Installer now uninstalls old eego versions	
Automatic naming of new sensor channels	3025
Calibration points for sensors now ordered in increasing voltage order	3026
Sensitivity selection for sensor channels with better default values	3119
Icon for video	2953
Icon for photic	2953
Review: removed video button in tab "playback control". If video review should be disabled, unselect it in Options -> Video -> Video review.	3003
Floating windows now have "re-dock" button instead of a close "[X]"	2955
Floating windows can now be controlled via the windows task bar.	
If there is no default set-up set, we choose the first hardware setup in the alphabetical list that has a "-CA-" substring as this defaults to standard waveguard caps	
The new montage dialog now uses the hardware setup selected on the Edit Montage workflow page. A separate selection of the hardware setup within the dialog is not possible any more.	
Renamed tabs: include words "eeg" and "online". Colors indicate the tab group.	2954, 2957
Review: renamed buttons Finalize / Cancel -> Restart / Finalize"	2969
Review: disabled scrollbar and render cursor interaction when playing video	2968
Zoom on selected button now resets the zoom to view all channels if no channels are selected.	
Eeg file export accepts unlimited number of time markers per segment now	2370
Warning message shown if export data format does not fully support data range	3128
The local database is not allowed to be the archive data base any more	3122
eego requires .Net 4.6 now	

Limitations

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Video recording properties like format, file size, resolution cannot yet be set.	2284
Playback speed of video cannot be changed while the video is playing. Press pause, change speed, press start again.	3124

Known problems

On rare occurances, rendering stops when starting a remote desktop connection. Recording will still continue in this case.	2104
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Very rarely, the amplifier cannot switch into impedance mode. In such a case, restart the amplifier and the recording.	2702
Archiving of recordings with averages not yet implemented.	2705
Channel selections from EEG preview mode are not synchronized with the channel selections in Impedance mode.	2756

Impedance values slightly unstable for caps with Sn electrodes.	2787
Triggercode is not correct if a bit is already high. Make sure to set all trigger inputs to low after a trigger.	2789
It is possible to save a Hardware Setup with all channels disabled	2805
256 channel layout for impedance slightly distorted	2815
Some sensor units exported to cnt (eeprobe format) are not available in asa 4.9.3.	3027
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Sometimes, video cannot be started. Workaround: reconnect camera and / or restart eego acquisition workflow.	3066, 3090
BrainVision export does not set reference channel label	3099
Sometimes, video shows unstable behaviour.	3131
Sometimes, a recording cannot be deleted because of a timeout of the database. Workaround: Start eego again.	3159

Version 1.6.2	Internal issue ID
Bug fixes since version 1.6.1	
Scale unit in the EEG view of the acquire WF reads uV	2717
Sensor data can not be scaled via the ribbon bar	2763
Some faulty recordings from old databases cannot be reviewed	2780
Trigger get lost when using cascaded system	2788
Sometimes, the review workflow cannot be started	2792
Handle missing / corrupt video files more gracefully: just don't play back video for them, but continue review. When reaching a part with normal video, play it back.	2793
Sometimes, eego doesn't start	2795
Synchronisation trigger forwarded in cascaded set-up with only 1 active amplifier	2798
Scrollbar runs away in review when cursor is at page boundary	2799
Sensor channels show values negative up.	2801
In acquire eeg: can proceed to recording step when the amplifier is switched off.	2804
Check for free size of backup device in archive	
Impedance view electrode position L1H in Duke 256 layout wrong	
Wrong display of elapsed time in review for recordings with multiple segments.	
Export to file fails due to missing dll	
Export to file fails for recordings with multiple segments	
Changes since version 1.6.1	
Improved impedance display for large channel counts	
Known problems	
Bipolar channels only support sampling rates up to 8196Hz. If a higher sampling rate is chosen, channels will deliver flat lines (0V).	2100
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Triggercode is not correct if a bit is already high. Make sure to set all trigger inputs to low after a trigger.	2789
It is possible to save a Hardware Setup with all channels disabled	2805

Version 1.6.1	Internal issue ID
New Features since version 1.5.6	
Support for amplifier sampling rates of up to 16384Hz .	
Cascaded recording with 2 or 4 EE-225 amplifiers to record 128 or 265 referential channels and 24 bipolar channels simultaneously with up to 16384Hz.	
Support for 4 sensor types: Temperature, Accelerometer, GSR, Respiration.	
Licensing scheme.	
Averaging (optional). The user can select a timespan around the trigger and a montage including filters. Average can be started, stopped, restarted, cleared. When recording, average is also recorded and is available in review.	
Power saving mode for mobile tablet usage, including automatic freeze of the rendering if user is inactive to save battery. For details, see the user guide.	
Monitoring of tablet or laptop battery state. Warn the user if battery gets too low. Savely stop the recording if battery state gets critical.	
Recovery for lost recordings. Note that it is only possible to restore finished segments.	2603
Offline file export into EEProbe format now also exports video and averages	
Extended data display in review. Show amplitudes at the time marker, impedances at segment start, clock time or elapsed time at the marker.	
Multiple video files can now be recorded within one EEG segment.	2307
Included audio recording in video, improved playback.	
Improved performance, reduced memory consumption.	
Added progress bar for database backup.	2733
Several improvements in EEG and trigger render display.	
Added more default amplifier set-ups and corresponding default montages: EEG + bipolar, EEG only, headbox XS-291.	
Added impedance layouts for all common 10-20-EEG cap and Duke cap layouts.	2656
Added application options to globally enable selection for Sampling Rate, Annotations, Video, and Photic.	

Changes since version 1.5.6

Default montages have now filters activated and channel lines colored.	2689
Export to file with filters and montage in review mode (offline mode) exports only the visible channels now.	2636
Archiving option now disabled by default.	2677
Restructured entries in application options.	
Restructured columns in Edit Amplifier table.	
Removed the "Up / Down" buttons in the Amplifier set-up.	2686
Removed button "accept changes" in the montage workflow. Press "Enter" instead or select another line.	2575
Option to export to file with or without applying filters is now avaiable directly in the workflow. Default can be set via application option.	2694
Impedance values now shown for all enabled referential channels, not only channels used in the current montage.	2667
Impedance layout re-organized	2720
Removed need to press control to zoom in impedance view	
Installer now installs codecs used in eego video recording so that other applications can play them back as well (e.g. asa).	2755
Averaging interval is now restricted to avoid OutOfMemory problems. The number of data points (= samples * channels) must be smaller than 180224.	
Updated database format to 7.0.8.	

Bug fixes since version 1.5.6

Incorrect battery state immediately after power on of the amplifier. Wait 5 seconds to settle.	1927
Event markers for video start not displayed during recording, but only in review.	2308
On rare occurences, video preview cannot be started. Workaround: Close and open eego, select camera anew in the options.	2378
On rare occurances, eego cannot be restarted, because a previous eego session didn't terminate properly. Workaround: restart the PC.	2401
On rare occurences, video capture cannot be started. Workaround: Stop the recording of the current segment, start video preview, start recording.	2406, 2436
On rare occurences, the user is not notified if a recording cannot be started due to an amplifier connection problem.	2428
Video does not support a disconnect / reconnect during the acquisition session. Workaround: stop acquisition and restart the workflow.	2431
On some tablet systems, the time cursors cannot be pulled to beginning and end of the screen.	2438
Starting a recording with a montage with no channels gives an exception	2455
Exception in acquisition when changing the display timespan with high frequency	2456
In acquire eeg: label of "stop impedance" not complete readable	2457
Hardware reference impedance label always visualized at CPz position.	2467
Gui artefacts switiching between video and event tab	2468
Impedance electrode visualization missing for inputs, solely used for rereferencing (bipolar references)	2480
In eeg export: events are not exported with brain vision	2489
Reference impedance is always zero	2497
Time cursor render displayed time off by 200ms for elapsed time	2515
Averaging with long timespan and high sampling rate can cause Out of Memory Exception	2516
OutOfMemory when exporting to file	2550
Exception in Downsampling during acquisition	2552
Copy/ paste functionality in montage workflow does not work	2571

Archiving of recordings with video is not possible	2587
Export to file sometimes failed	2595
Switching the amplifier type takes very long	2601
Data is lost when tablet (8inch) turns off and no subject was selected	2603
Exception in filtering: Pole unstable	2611
No error message given if streaming cannot be started	2613
Acquisition workflow hangs when disconnect during recording start	2614
Error when deleting channel in the montage that was just created	2618
Brainvision file export fails	2619
Video cannot be started after "Stop Video"-Button was clicked	2623
Data from older databases with amplifier setups that are not existing any longer can't be exported with a related montage	2635
Exported cnt with channels with a space in the name cannot be opened in asa	2638
Cannot delete recordings with average entry	2664
Visualization flag not set for old montages	2672
Impedance values reset in rendering when changing the montage	2673
Annotation are not set while freeze but is active	2679
Video: Timeouts and StateChangeNotAllowed errors	2683
Crash starting review - FindImpedanceValueForMontageChannel	2684
In edit montage: cancel when deleting a montage jumps back to default montage	2687
Incorrect tooltip in montage when error in filter cutoff frequency	2700
MainWindow fades back before last pop up is closed	2701
Out of memory exception with 128 channels and display span 4s	2703
After exception, videos are not saved in the correct way and the whole file can't be opened in the review	2704
Average ribbon does not update the trigger count	2710
Export of average within the marker doesn't work	2711
Trigger display in render off by one sample	2730
Video is not exported when intervals via S1 and s2 are used	2732
There is no check in backup to check whether the external drive has enough space to store the database	2735
No warning that backup is in progress when closing the application	2736
Backup fails if database name contains a space	2746
Known problems	
Bipolar channels only support sampling rates up to 8196Hz. If a higher sampling rate is chosen, channels will deliver flat lines (0V).	2100
On rare occurances, rendering stops when starting a remote desktop connection. Recording will still continue in this case.	2104
Eego freezes if connection with database is interrupted while recording.	2271
Data is lost if connection with database is interrupted while recording. There are only extreme cases that lead to a loss of database connection, e.g. forcefully kill the eego application. For extended safety, users might enable the cnt export to file during acquisition.	2272
Video recording properties like format, file size, resolution cannot yet be set.	2284
Duration of video recording not yet visualized in review.	2311
In a video session with 1 monitor after a video session with two monitors, the undocked video window can be out of the bounds of the monitor.	2396
Montage editor: move ref channel past bipolar channel duplicates bipolar channel in view	2499

After a restart of the computer, very rarely no data are delivered from the amplifer because of a USB error. In such a case, disconnect and re-connect the USB cable.	2612
After an error, sometimes the amplifer state or battery state are displayed incorrectly in the status bar. Restarting eego and the amplifier helps.	2624, 2625
Very rarely, the amplifier cannot switch into impedance mode. In such a case, restart the amplifier and the recording.	2702
Archiving of recordings with averages not yet implemented.	2705
Channel selections from EEG preview mode are not synchronized with the channel selections in Impedance mode.	2756
Sensor data can not be scaled via the ribbon bar	2763
On rare occurances, eego might fail with an OutOfMemory exception. The software tries to change the configuration in that case to minimize memory consumption in trade off for computation speed.	2774, 2775
Sometimes, USB detach of a video camera during video recording causes eego to shut down.	2776
When creating a montage as a copy of another montage, not all properties of the template are kept.	2778

Version 1.5.6	Internal issue ID

New Features since version 1.5.5

"Freeze" button in EEG acquisition ribbon will now disable all computations that are only required for displaying the EEG. Thus, CPU power will be saved, allowing mobile computers like tablets to run at low CPU speed, saving battery and preventing thermal stress.

Changes since version 1.5.5

eego software now monitors the delay between data acquisition and display of the data. If the delay is bigger than 15 seconds, it is an indication that the CPU performance is insufficient. In this case, the acquisition workflow is stopped automatically. The recorded data up to this point will be saved.

If the freeze button is pressed, the Annotation buttons get disabled.

Bug fixes since version 1.5.5

Datastreaming doesn't recover from OutOfMemory: eego freezes, recording lost	2524
Unresponsive GUI and long delay in rendering if tablets get hot	2537
Wrong channel data if channels disabled in amplifer set-up	2545
No impedance data persisted	2546
Impedances on wrong channel if channels disabled in amplifer set-up	2547

Known problems

see version 1.5.4

Version 1.5.5	Internal issue ID

Bug fixes since version 1.5.4

Delay in the display of rendered data after long recordings	2478
Faulty default montage for amplifier EE-201 and amplifier setup EEG32_EMG16_CA_200_4.	2506
Increased timeout value for EE-201, EE-202, EE-203, EE-206, EE-207 amplifiers to increase stability	2512

Known problems

see version 1.5.4

Version 1.5.4

New Features since version 1.5.3

"Play mode" in review, running through the complete EEG incuding multiple segments and video files

Layout improvements in tablet and desktop mode	2053, 2407, 2419, 2440
Export of video and trigger data when exporting multiple recordings	2433
Increased responsiveness of the recording, i.e. reduced the latency between signal input into the amplifier and display of the signal on screen.	
Option to disable the "success" message after export to file	
Changes since version 1.5.3	
Review: Selection of event in event grid should focus eeg	2056
FAT32 devices are no longer supported for database backup as they only allow 4GB of storage.	2429
Channel selection combobox allows now also 80 channels	2442
Display timespan during recording has additional values of 2 seconds and 1 second.	
Software operates on .Net version 4.5.1 now	

Bug fixes since version 1.5.3	Internal issue ID
In rare occurences, triggers during impedance were delivered with the next chunk of EEG data	2106
Video play backwards and high speed playback not working	2300
On rare occurances during long term recording with high memory and CPU strain, memory cannot be freed as fast as new data comes in, leading to an OutOfMemory error. Workaround: Increase the display timespan, so that the screen refreshes less often.	2393
Exception in FilterBuffer during long term recording	2395
Can't review data, when the video recording was interrupted by disconneting the camera	2416
Out of memory exception in review for 0.01 s display time span	2426
Wrong time displayed in ribbon for time cursor	2427
No error message when video camera is disconnected during preview	2430
Error exporting files if the filename already exists	2418, 2432
Disk space not freed directly after deleting recordings	2435
Wrong video count if video connection is interupted	2437
Display problems for freeze frame in video review	2445
Direct export doesn't work with the default path settings	2446
Direct export during recording not possible if no subject is selected	2447
Several issues with video playback	2449
In rare occurences, a trigger was split into two triggers on subsequent samples; the sum of the two new trigger codes matching the correct trigger code.	2450
Exception when deleting a subject	2453
Exception at the end of the acquisition session	2454
Subject cannot be selected, workflow stops	2458

Known problems	Internal issue ID
Incorrect battery state immediately after power on of the amplifier. Wait 5 seconds to settle.	1927
Bipolar channels only support sampling rates up to 8196Hz. If a higher sampling rate is chosen, channels will deliver flat lines (0V).	2100
Eego freezes if connection with database is interrupted while recording.	2271
Data is lost if connection with database is interrupted while recording. There are only extreme cases that lead to a loss of database connection, e.g. forcefully kill the eego application. For extended safety, users might enable the cnt export to file during acquisition.	2272
Video recording properties like format, file size, resolution cannot yet be set.	2284
Recording of multiple video files per segment not yet possible.	2307

Event markers for video start not displayed during recording, but only in review.	2308
Duration of video recording not yet visualized in review.	2311
On rare occurences, video preview cannot be started. Workaround: Close and open eego, select camera anew in the options.	2378
On rare occurances, eego cannot be restarted, because a previous eego session didn't terminate properly. Workaround: restart the PC.	2401
On rare occurences, video capture cannot be started. Workaround: Stop the recording of the current segment, start video preview, start recording.	2406, 2436
On rare occurences, the user is not notified if a recording cannot be started due to an amplifier connection problem.	2428
Video does not support a disconnect / reconnect during the acquisition session. Workaround: stop acquisition and restart the workflow.	2431
On some tablet systems, the time cursors cannot be pulled to beginning and end of the screen.	2438
Starting a recording with a montage with no channels gives an exception	2455
Exception in acquisition when changing the display timespan with high frequency	2456

Version 1.5.3

New Features since version 1.2.1

Additional sample rates added from 4000 Hz up to 16384 Hz (depending on amplifier type)

Support for new amplifier types added: EE_205, EE-206, EE-207, EE-208, EE-209, EE-211, EE-212, EE-213, EE-214, EE-215, EE-221, EE-222, EE-223, EE-224, EE-225

Review of recorded or archived eeg from within eego. Data is displayed with montage applied, i.e. it can be re-referenced and filtered for display. Export of re-referenced and filtered data to file is supported. Export of portions of the recording to file are supported.

Added table "data" in review, displaying amplitudes and amplitude differences for two selected time points for all channels. Montage with filters and re-referencing is applied.
Export of recordings to file. Different file formats are supported: extended cnt format (eeprobe64 bit), edf, Neuroscan cnt, BrainVision eeg, ASCII. Recordings can either be exported as recorded, or filtered and re-referenced by a montage. Export of multiple recordings in one operation is supported.
Direct export of recordings to extended cnt format (eeprobe 64bit) while recording. Feature can be enabled via Options -> Storage.

Montage allows common average of all EEG channels as reference.

Support for Windows 8.x-Tablets added. Eego GUI can be switched into tablet mode via Options -> General.

Improved performance, resulting in lower CPU strain and thus extended notebook battery times.

Enable / disable channels in amplifier setup

Amplifier type can now be adjusted in the Options

Added wait cursor while deleting recordings in Edit Subject workflow

Added delete button for subjects and recordings in edit subject workflow

Extended status bar information: recording frequency

Display option: show zero lines

Review in ASA respects sub-millisecond timing now as introduced with sampling frequencies up to 16kHz.

Experimental Feature: Video recording and video review. Contact your distributor or eemagine support for system requirements and eligibility.

Experimental Feature: Photic stimulation using the eelite device. Contact your distributor or eemagine support for system requirements and eligibility.

Experimental Feature: Support for combined recording with two amplifiers added. Contact your distributor or eemagine support for system requirements and eligibility.

Changes since version 1.2.1

Updated database format

Moved "Use Clock Time" and "Switch Polarity" to Options. These values need now to be set before the session.

Changed control for Default Sampling Rate form RadioButton to ComboBox

Render controls: Extended sensitivity value maximum to 1V for referential channels

Warning for low battery level now appears at 25% (when the battery indicator at the device shows a red light). Amplifier turns off at 15% battery level to prevent complete draining and damage of the battery.

Default amplifier setups and montages adjusted. By default, the maximum number of channels for a given amplifier type is enabled. Signal range for referential channels is 150mV, signal range for bipolar channels is 350mV. The amplifier setup is by default linked to the standard waveguard cap that is shipped with the amplifier. The name of the amplifier setup reflects the waveguard product code and revision.

Display timespan during recorded has now 4 seconds as minimum timespan to avoid too high performance strain on tablet CPUs.

Bug fixes (including fixes for 1.4.x and 1.5.x beta versions)	Internal issue ID
Fullscreen of eeg window hides the windows taskbar	946
Case sensitive name of user yields confusing error message creating a new user	1051
After adding around 100 manually triggers, the workflow toolbar becomes inactive	1243
Large string for channel name is cut off in recording page	1263
Amplifier disconnection in Impedance view leads to a buggy state of the recording workflow.	1324
GND channel is not displayed in impedance view	1332
Subjects cannot be deleted in error state	1350
Exception in Edit subject workflow when two subjects have the same name and one of them has a recording	1351
EEG is not loaded for second database	1389
Exception when click on annotations via remote desktop	1406
Cannot set a montage setup as default montage	1442
Exception when deleting channels in new montage	1443
Change montage in acquire eeg workflow does not update frequency values in main ribbon tab	1446
Steepness and notch cannot be modified in group in montage editor	1679
Channel labels are cropped, if name too long	1729
Editing of channels in recording workflow is not done for all selected channel	1741
Annotations are not synchronized between workflow and general application options	1779
Asa import: triggers rounded to millisecond	1790
Filter: Wrong scaling for steepness 6, 18 and 30 for signal of 15 Hz with cut off frequency of 15Hz	1812
Hardware Template is the default hardware setup and not the one selected in the menu	1836
GUI in Analyze EEG workflow should not allow the user to select more than one recording	1837
Electrode layout in Impedance mode	1860
CPz reference label remains gray colored during impedance measurement.	1873
Disabled insertion of annotations if annotation text is invalid (e.g. empty)	1880
Recording cannot be started if the application was started while the amplifier was powered off	1926
After power off of the amplifier, there might be still a battery level displayed in the application. Amplifier status is presented correctly.	1951
In rare occurrences, the hardware freezes when changing between recording and impedance modes. In this case, an error message is displayed and any recorded data can be saved. Turn off the device, disconnect USB, turn the device on again and re-connect USB to be able to start a new recording.	1957

In rare occurrences, recording cannot be started due to incorrect state of the amplifier after a previous session. In this case, an error message is displayed. Turn off the device, disconnect USB, turn the device on again and re-connect USB to be able to start a new recording.	1966
Random or zero impedance values might be displayed if channel is open	1996
Impedance view electrode layout is incorrect when only a few channels are active.	1998
For recording to file, the name of the exported data does not contain the name of the subject if user proceeds to recording page without selecting the patient.	2212
Zoom on selected doesn't ignore invisible channels.	2248
Impedance values not available from recording to cnt file.	2266
Sometimes video preview or video recording cannot be started. If it fails, close eego, disconnect and reconnect camera and stat over again.	2319
Scaling by mouse up/down in acquisition leads to filter settling artifacts	2333
Sometimes the amplifier not in Idle mode after fast finalizing a record after stop record, preventing a second start of the acquisition. Workaround: quit and restart eego.	2360
Zooming and shifting the video in ist window has limited functionality	2367

Known problems	Internal issue ID
Incorrect battery state immediately after power on of the amplifier. Wait 5 seconds to settle.	1927
Bipolar channels only support sampling rates up to 8196Hz. If a higher sampling rate is chosen, channels will deliver flat lines (0V).	2100
Eego freezes if connection with database is interrupted while recording.	2271
Data is lost if connection with database is interrupted while recording. There are only extreme cases that lead to a loss of database connection, e.g. forcefully kill the eego application. For extended safety, users might enable the cnt export to file during acquisition.	2272
Video recording properties like format, file size, resolution cannot yet be set.	2284
Video playback "backwards" and playback with speed more than 2x is not yet possible.	2300
Recording of multiple video files per segment not yet possible.	2307
Event markers for video start not displayed during recording, but only in review.	2308
Duration of video recording not yet visualized in review.	2311
On rare occurences, video preview cannot be started. Workaround: Close and open eego, select camera anew in the options.	2378
On rare occurances during long term recording with high memory and CPU strain, memory cannot be freed as fast as new data comes in, leading to an OutOfMemory error. Workaround: Increase the display timespan, so that the screen refreshes less often.	2393
On rare occurances, eego cannot be restarted, because a previous eego session didn't terminate properly. Workaround: restart the PC.	2401
If a recording with video terminates due to an internal error (exception), it might not be possible to review the recording in eego.	2416

Version 1.2.1

Build 2.1.1043.20385

New Features

Start and Stop Recording on Amplifier Triggers

Known problems	Internal issue ID
Adding annotations with very high frequency might freeze the recording view. Limit entry of annotations to one key stroke per second.	1243
Very high input trigger rates in the range of 10kHz and way larger than the sampling rate may cause the status bar to freeze. Restrict input trigger rate to meaningful values in the range of the sampling rate.	1320

If connection to the archive server is interrupted, eego™ the software takes a while until this is recognized and will slow down.	1309
Connection of several eegosport systems to a single archive server at the same time can lead to blocking of ultrabooks while one of the systems is archiving. Connecting several eegosports ultrabooks at the same time is not a supported use case for the current version.	1198
Archiving on a server with no free space appears in the eego application like an error with no explanatory message.	1196
Montage with only 24 bipolar channels is missing.	1330
Bipolar channels wrongly display impedance values. Impedance is not measured for these channels and displayed values will not be correct.	1331
GND channel is not displayed in impedance view.	1332

Bug fixes

The row IDs in the amplifier setup grid views are missing.	1327
The application option allowing to set the maximum number of segments for a recording session has no upper and no lower limit. Should not be changed from the default value 256 without very good reason.	1328
Visibility check box for recordings in “Edit Subject” workflow cannot be unchecked.	1334

Version 1.2.0

Build 2.0.1023.20115

New Features

New amplifier type included: eegosportsTM EE-203 revision 1.0.

New amplifier type included: eegosportsTM EE-202 revision 6.0.

Annotations can be added to the EEG recording (as trigger information with dedicated codes).

Archiving workflow added for moving data to the archive server.

Archive server data can be accessed from the eego system in the same way as data in the local database.

Backup functionality added to create safety copy of database.

Bipolar channels in addition to referential channels can be recorded and reviewed.

Archive server XS-301 is added to the system.

Known problems	Internal issue ID
Adding annotations with very high frequency might freeze the recording view. Limit entry of annotations to one key stroke per second.	1243
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Montage with only 24 bipolar channels is missing.	1330

Bipolar channels wrongly display impedance values. Impedance is not measured for these channels and displayed values will not be correct.	1331
GND channel is not displayed in impedance view.	1332
Visibility check box for recordings in “Edit Subject” workflow cannot be unchecked.	1334

Version 1.1.1

Build 1.1.565.17158

Known problems	Internal Problem ID
When recording and using impedance check, first ten seconds of triggers are lost.	954
Sometimes, under unknown circumstances eego software has been seen to slow down or freeze.	964
Sometimes, under unknown circumstances, CPU load can raise to levels twice as high as usual. This doesn't affect normal behavior of the software.	933
After changing the number of displayed channels in the EEG display, channels are not centered vertically.	973
A switched off amplifier that's connected to the software can be selected as recording device.	493
Recording more than ten thousand triggers will slow down **ee**go™ dramatically when finalizing the recording workflow.	983